

Python and Machine Learning

Seminar 1: Introduction and Python

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Welcome!

- Who am I?
 - Adjunct Instructor in the Department of Economics
 - Research in Behavioral Economics and Applied Microeconometrics
- **January 23-24-25.** 9h00-16h00
- **20/2, 27/3 and 25/4.** 19h15-21h15
- Contact: emirtarik.dakin@sciencespo.fr or you can book an appointment.

- **Course material:** seminar slides will be uploaded on Moodle, along with readings and other supplementary material.
- This class is an introductory Python course. However, it also contains new concepts of applied econometrics and statistical theory.
 - Try to recap everything we do in class, both theory and practice. Most of these concepts or even your own writing/code will look alien to you after a week if you don't review them.
 - Work with each other.
 - Ask questions, clarify doubts as soon as you have them.
- You are **encouraged to use ChatGPT** for coding and understanding concepts. However, it doesn't make sense to ask everything from ChatGPT as you'll soon understand that you're not actually learning anything while trying to make things easier.
- You will get a **zero** for assignments that use AI text (not code), not to mention a possible disciplinary action for plagiarism. Please be careful.

Assessment and grading

There are **two** main assessments for your total grade.

❶ **Presentations** (60%).

- In groups of four.
- You will present your work in short sessions (about 10 minutes) which will be followed by questions.
- There will be three presentation sessions (worth 20% each) for each group.

❷ **Report** (40%).

- Same groups.
- At the end of the semester, you will turn in a short replica of an academic paper, which will include your story, data/methodology and your findings.
- **Presentations** - Thursday sessions.
- **Paper deadline** - May 9th.

- This class is mainly about Python and Machine Learning but these are only tools. We will use these tools for applied topics, starting off with simple regression analysis and moving onto more sophisticated methods for observational data.
- By **observational data**, we mean situations in which the information was collected without manipulation from an investigator. In other words, no experiments involved.
- The main issue with observational data is that sometimes we want to answer research questions that are about **cause and effect**.
- And answering cause and effects questions with observational data is not always straightforward, because it has to be tailored to our needs.
- Observational data is often cluttered with information, which makes it hard to see what's really relevant—**curse of dimensionality**.

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- This is an **applied** methods class. We won't ignore statistical theory but there is no need to be proving theorems.
- Understanding theory beyond the proofs is the goal of this class. To be a good applied empirical researcher you need to understand different methods and know what theory tells you about them.
- The statistics —and econometrics and machine learning as eventually follows— revolution of the last 40 years has been the result of raw computing power. This is where Python comes into play.

Why regression models?

- There are several reasons to run a regression:
 - **describing** a relationship
 - making **causal** statements and testing hypotheses
 - making **predictions**
- The purpose of running a model changes the way you think about **assumptions and the way you conduct the analysis**. We will often come back to the term **causal inference** when reading papers and analyzing data.
- Anybody can analyze randomized data (run experiments); working with **observational data** is a lot harder.

- Regression models are key to drawing **causal inferences** from observational data.
- We will start by **defining causal effects** and introducing some new language.
- We will review regression results focusing on interpretation and practical issues.
- Then we will talk about **research designs** and their assumptions without using statistical models. You will be thinking about datasets and ways in which you could apply these methods.

Using causal inference

- Does immigration lead to lower wages and/or higher unemployment for locals?
- Does raising the minimum wage reduce employment for low-skilled labor?
- How much effect of an additional year of higher education have on wages?
- Do higher public debt levels lead to lower economic growth?
- Does the neighborhood you grow up in have an effect on your life outcome?

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Research Designs

- By **research design**, we mean the overall strategy used to answer a research question.
- This is different than the **estimation** of effects or the particular regression model or statistical approach we use to answer a research question.
- Traditional econometrics (as in Wooldridge) tends to combine **design** and **estimation**, but that might be more confusing at times.
- For example, whether causality is established or not is discussed in terms of assumptions about the error term $\epsilon \sim N(0, \sigma^2)$ (i.e., **conditional independence assumption**).
- But then in other models like logistic, probit, or Poisson, there is no additive error term and so the intuition of conditional independence gets you tied in knots.
- Angrist and Pischke (2009) is a decent bridge (*Mostly Harmless Econometrics*). For a more intuitive approach, you can take a look at the book *Mastering 'Metrics*, which is also by Angrist and Pischke (2014).

Why Machine Learning?

- Machine Learning (ML) offers methods that complement traditional econometric techniques:
 - **Handling complex patterns:** ML models can capture non-linearities and interactions in data that basic econometric models may miss.
 - **High-dimensional data:** Effective in settings with many predictors or features, where traditional models can struggle.
 - **Automated model selection:** Techniques like cross-validation and ensemble methods help in choosing models with better predictive performance.
- ML emphasizes **prediction accuracy** and **data-driven decision making**, providing powerful tools for forecasting and classification tasks.
- While econometrics focuses on causal inference and hypothesis testing, ML expands our toolkit for **prediction** and **pattern discovery** in complex datasets.

From data analysis to data science

- **Data analysis** (or **data mining**): subfield of statistics and computer science. Statistical methods aimed at summarizing the information contained in databases with the goal of extracting knowledge. Data analysts make sense of the data for the decision-makers who need to use it.
- Each method is adapted to a given type of data and to a precise objective. These methods are the basic building blocks to understanding the **data science** field.
 - data visualization
 - dimension reduction — PCA
 - segmentation — clustering
 - discriminant analysis and decision trees
- **Objective**: presenting some of these statistical methods with some basics for analyzing large datasets.

Two types of methods

- ① **Unsupervised learning:** finding patterns in data without any prior knowledge of the structure of the data.
 - no output variable
 - no training set
 - PCA and clustering methods
 - **Examples:** customer segmentation, anomaly detection, dimension reduction

- ② **Supervised learning:** predicting the value of an output variable based on the values of input variables.
 - labeled training set (with output)

 - ① **Classification:** output variable is a category.
 - **Example:** predicting whether an email is spam or not.

 - ② **Regression:** output variable is a real number.
 - **Example:** predicting the price of a house based on its characteristics.

Statistical units and variables

- **Unit** (or observation or case): basic element in statistics (students, firms, countries, years etc.).
- **Variable**: characteristic measured on the units (gender, age, number of employees, GDP, unemployment rate etc.).
 - variables can be **categorical** or **numerical**.

Univariate analysis

- **Graphs:**

- categorical variable: pie chart or bar chart
- numerical variable: histogram, boxplot

- **Numerical indicators:** characteristic measured on the units (gender, age, number of employees, GDP, unemployment rate etc.).

- categorical variable: mode or calculus on frequencies
- numerical variable (x): mean, variance, standard deviation

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$\text{Var}(x) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

$$\sigma_x = \sqrt{\text{Var}(x)}$$

Check out more graphs [here](#).

- **Graphs:**

- 2 categorical variables: stacked or juxtaposed bar charts
- 2 numerical variables: scatterplot
- 1 numerical and 1 categorical variable: parallel boxplots

- **Numerical indicators:** characteristic measured on the units (gender, age, number of employees, GDP, unemployment rate etc.).

- categorical variable: independence chi-square
- numerical variable (x and y): covariance and correlation coefficient

$$\text{Cov}(x, y) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x}) (y_i - \bar{y})$$
$$\rho(x, y) = \frac{\text{cov}(x, y)}{\sigma_x \sigma_y}$$

Our statistical tool: Python

- Python is a high-level programming language that is widely used in academia and industry.
- Renowned for its readability, ease of learning, and vast ecosystem.
- In social sciences, we mostly use R as it's generally sufficient for econometrics.
- But for more advanced tools like machine learning, data science and quantitative economics, we rely on Python's well-established ecosystem.

- AI is already remarkably good at helping you write code.
 - We will integrate working with AI with our learning of Python.
- Python is playing a huge role in the advance of AI and machine learning.
 - Tech firms are pouring money into development of extremely powerful Python libraries.
- Even if you don't plan to work on AI and machine learning, you can benefit from learning to use some of these libraries for your own projects in economics and other fields of science.

Getting Started with Python

We will use Google's **Colab** to learn Python.

- it doesn't require any setup.
- easy to work together on (like Google docs).

To install Python locally (optional):

- Install Python from the official website (<https://www.python.org/>).
- Use package managers like **pip** or **conda** to manage libraries.
- Explore Integrated Development Environments (IDEs) such as VS Code (recommended), PyCharm, or Jupyter Notebook.
- Take advantage of free online resources and communities.

So, let's switch over to Python.